NAME

latexmk − generate LaTeX document

SYNOPSIS

latexmk [options] [file ...]

DESCRIPTION

Latexmk completely automates the process of compiling a LaTeX document. Essentially, it is like a specialized relative of the general *make* utility, but one which determines dependencies automatically and has some other very useful features. In its basic mode of operation *latexmk* is given the name of the primary source file for a document, and it issues the appropriate sequence of commands to generate a .dvi, .ps, .pdf and/or hardcopy version of the document.

By default *latexmk* will run the commands necessary to generate a .dvi file, which copies the behavior of earlier versions when only *latex* was available.

Latexmk can also be set to run continuously with a suitable previewer. In that case the*late x* program (or one of its relatives), etc, are rerun whenever one of the source files is modified, and the previewer automatically updates the on-screen view of the compiled document.

Latexmk determines which are the source files by examining the log file. (Optionally, it also examines the list of input and output files generated by the **-recorder** option of modern versions of *latex* (and *pdflatex*, *xelatex*, *lualatex*). See the documentation for the**-r ecorder** option of *latexmk* below.) When*late xmk* is run, it examines properties of the source files, and if any have been changed since the last document generation, *latexmk* will run the various LaTeX processing programs as necessary. In particular, it will repeat the run of *latex* (or a related program)) often enough to resolve all cross references; depending on the macro packages used. With some macro packages and document classes, four, or even more, runs may be needed. If necessary, *latexmk* will also run *bibtex*, *biber*, and/or *makeindex*. In addition,*late xmk* can be configured to generate other necessary files. For example, from an updated figure file it can automatically generate a file in encapsulated postscript or another suitable format for reading by LaTeX.

Latexmk has two different previewing options. With the simple **-pv** option, a dvi, postscript or pdf previewer is automatically run after generating the dvi, postscript or pdf version of the document. The type of file to view is selected according to configuration settings and command line options.

The second previewing option is the powerful **-pvc** option (mnemonic: "preview continuously"). In this case, *latexmk* runs continuously, regularly monitoring all the source files to see if any have changed. Every time a change is detected, *latexmk* runs all the programs necessary to generate a new version of the document. A good previewer will then automatically update its display. Thus the user can simply edit a file and, when the changes are written to disk, *latexmk* completely automates the cycle of updating the .dvi (and/or the .ps and .pdf) file, and refreshing the previewer's display. It's not quite WYSIWYG, but usefully close.

For other previewers, the user may have to manually make the previewer update its display, which can be (e.g., with some versions of *xdvi* and *gsview*) as simple as forcing a redraw of its display.

Latexmk has the ability to print a banner in gray diagonally across each page when making the postscript file. It can also, if needed, call an external program to do other postprocessing on generated dvi and postscript files. (See the options **-dF** and **-pF**, and the documentation for the *$dvi_filter* and *$ps_filter* configuration variables.) These capabilities are leftover from older versions of *latexmk*, but **are currently nonfunctional**. More flexibility can be obtained in current versions, since the command strings for running **latex* can now be configured to run multiple commands. This also extends the possibility of postprocessing generated files.

Latexmk is highly configurable, both from the command line and in configuration files, so that it can accommodate a wide variety of user needs and system configurations. Default values are set according to the operating system, so *latexmk* often works without special configuration on MS-Windows, cygwin, Linux, macOS, and other UNIX systems. See the section "Configuration/Initialization (rc) Files", and then the later sections "How to Set Variables in Initialization Files", "Format of Command Specifications", "List of Configuration Variables Usable in Initialization Files", "Custom Dependencies", and "Advanced